C1 Equation of a straight line

Name: ____________________

Class: ____________________

Date: ____________________

Time: **388 minutes**

Marks: **325 marks**

Comments:

Q1.

Three of the following points lie on the same straight line.

Which point does **not** lie on this line?

Tick **one** box.

(−2, 14)	☐
(−1, 8)	☐
(1, −1)	☐
(2, −6)	☐

(Total 1 mark)

Q2.

A circle has equation $(x - 2)^2 + (y + 3)^2 = 13$

Find the gradient of the tangent to this circle at the origin.

Circle your answer.

$-\frac{3}{2}$ $\qquad -\frac{2}{3}$ $\qquad \frac{2}{3}$ $\qquad \frac{3}{2}$

(Total 1 mark)

Q3.

Point *C* has coordinates $(c, 2)$ and point *D* has coordinates $(6, d)$.

The line $y + 4x = 11$ is the perpendicular bisector of *CD*.

Find c and d.

(Total 5 marks)

Q4.

Points *A* (−7, −7), *B* (8, −1), *C* (4, 9) and *D* (−11, 3) are the vertices of a quadrilateral *ABCD*.

(a) Prove that *ABCD* is a rectangle.

(4)

(b) Find the area of *ABCD*.

(2)
(Total 6 marks)

Q5.

Three points *A*, *B* and *C* have coordinates *A* (8, 17), *B* (15, 10) and *C* (−2, −7)

(a) Show that angle *ABC* is a right angle.

(3)

(b) *A*, *B* and *C* lie on a circle.

(i) Explain why *AC* is a diameter of the circle.

(1)

(ii) Determine whether the point *D* (−8, −2) lies inside the circle, on the circle or outside the circle.

Fully justify your answer.

(4)
(Total 8 marks)

Q6.

The line *L* has equation $2x + 3y = 7$

Which one of the following is perpendicular to *L*?

Tick **one** box.

$2x - 3y = 7$ ☐

$3x + 2y = -7$ ☐

$2x + 3y = -\frac{1}{7}$ ☐

$3x - 2y = 7$ ☐

(Total 1 mark)

Q7.

Determine whether the line with equation $2x + 3y + 4 = 0$ is parallel to the line through the points with coordinates (9, 4) and (3, 8).

(Total 4 marks)

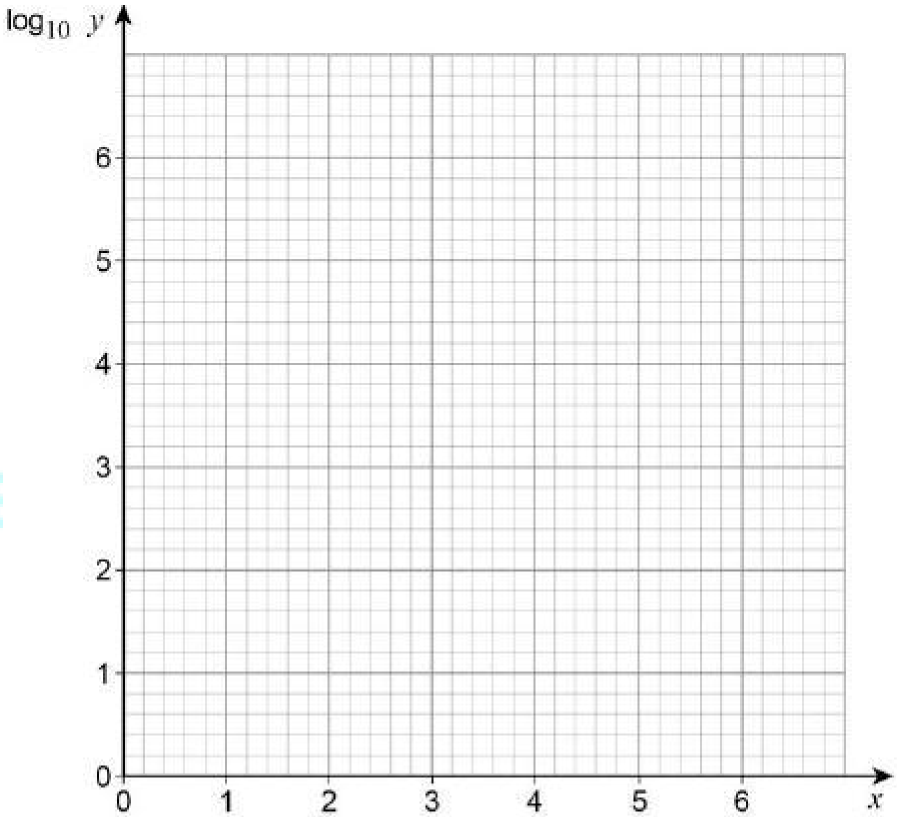
Q8.

A student conducts an experiment and records the following data for two variables, x and y.

x	1	2	3	4	5	6
y	14	45	130	1100	1300	3400
$\log_{10} y$						

The student is told that the relationship between x and y can be modelled by an equation of the form $y = kb^x$

(a) Plot values of $\log_{10} y$ against x on the grid below.



(2)

(b) State, with a reason, which value of y is likely to have been recorded incorrectly.

(1)

(c) By drawing an appropriate straight line, find the values of k and b.

(4)

(Total 7 marks)

Q9.

A curve has equation $y = 6x\sqrt{x} + \dfrac{32}{x}$ for $x > 0$

(a) Find $\dfrac{dy}{dx}$

(4)

(b) The point A lies on the curve and has x-coordinate 4

Find the coordinates of the point where the tangent to the curve at A crosses the x-axis.

(5)
(Total 9 marks)

Q10.

The circle with equation $(x - 7)^2 + (y + 2)^2 = 5$ has centre C.

(a) (i) Write down the radius of the circle.

(1)

(ii) Write down the coordinates of C.

(1)

(b) The point P (5, −1) lies on the circle.

Find the equation of the tangent to the circle at P, giving your answer in the form $y = mx + c$

(4)

(c) The point Q (3, 3) lies outside the circle and the point T lies on the circle such that QT is a tangent to the circle. Find the length of QT.

(4)
(Total 10 marks)

Q11.

Find the gradient of the line with equation $2x + 5y = 7$

Circle your answer.

$\dfrac{2}{5}$ $\qquad$ $\dfrac{5}{2}$ $\qquad$ $-\dfrac{2}{5}$ $\qquad$ $-\dfrac{5}{2}$

(Total 1 mark)

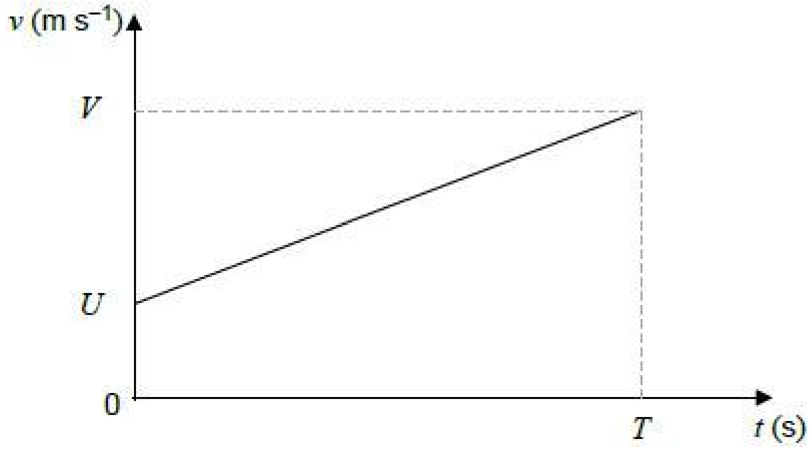
Q12.

A particle moves on a straight line with a constant acceleration, a m s $^{-2}$.

The initial velocity of the particle is U m s $^{-1}$

After T seconds the particle has velocity V m s $^{-1}$

This information is shown on the velocity-time graph.



The displacement, S metres, of the particle from its initial position at time T seconds is given by the formula

$$S = \frac{1}{2}(U + V)T$$

(a) By considering the gradient of the graph, or otherwise, write down a formula for a in terms of U, V and T.

(1)

(b) Hence show that $V^2 = U^2 + 2aS$.

(3)

(Total 4 marks)

Q13.

The point A has coordinates (−3, 2) and the point B has coordinates (7, k).

The line AB has equation $3x + 5y = 1$.

(a) (i) Show that $k = -4$.

(1)

(ii) Hence find the coordinates of the midpoint of AB.

(2)

(b) Find the gradient of AB.

(2)

(c) A line which passes through the point A is perpendicular to the line AB. Find an equation of this line, giving your answer in the form $px + qy + r = 0$, where p, q and r are integers.

(3)

(d) The line AB, with equation $3x + 5y = 1$, intersects the line $5x + 8y = 4$ at the point C. Find the coordinates of C.

(3)
(Total 11 marks)

Q14.

(a) A curve is defined by the equation $x^2 - y^2 = 8$.

(i) Show that at any point (p, q) on the curve, where $q \neq 0$, the gradient of the curve is given by $\frac{dy}{dx} = \frac{p}{q}$.

(2)

(ii) Show that the tangents at the points (p, q) and $(p, -q)$ intersect on the x-axis.

(4)

(b) Show that $x = t + \frac{2}{t}$, $y = t - \frac{2}{t}$ are parametric equations of the curve $x^2 - y^2 = 8$.

(2)
(Total 8 marks)

Q15.

The line AB has equation $3x - 4y + 5 = 0$.

(a) The point with coordinates $(p, p + 2)$ lies on the line AB. Find the value of the constant p.

(2)

(b) Find the gradient of AB.

(2)

(c) The point A has coordinates $(1, 2)$. The point $C(-5, k)$ is such that AC is perpendicular to AB. Find the value of k.

(3)

(d) The line AB intersects the line with equation $2x - 5y = 6$ at the point D. Find the coordinates of D.

(3)
(Total 10 marks)

Q16.

A curve has equation $y = x^5 - 2x^2 + 9$. The point P with coordinates $(-1, 6)$ lies on the curve.

(a) Find the equation of the tangent to the curve at the point P, giving your answer in the form $y = mx + c$.

(5)

(b) The point Q with coordinates $(2, k)$ lies on the curve.

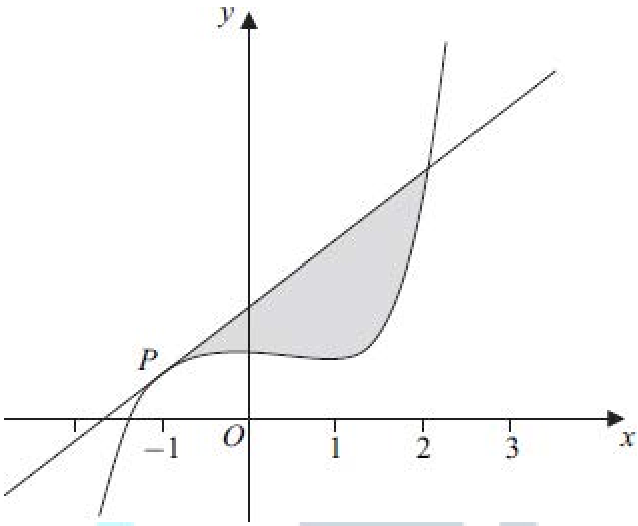
(i) Find the value of k.

(1)

(ii) Verify that Q also lies on the tangent to the curve at the point P.

(1)

(c) The curve and the tangent to the curve at P are sketched below.



(i) Find $\int_{-1}^{2} (x^5 - 2x^2 + 9)\,dx$.

(5)

(ii) Hence find the area of the shaded region bounded by the curve and the tangent to the curve at P.

(3)

(Total 15 marks)

Q17.

The point A has coordinates $(6, -4)$ and the point B has coordinates $(-2, 7)$.

(a) Given that the point O has coordinates $(0, 0)$, show that the length of OA is less than the length of OB.

(3)

(b) (i) Find the gradient of AB.

(2)

(ii) Find an equation of the line AB in the form $px + qy = r$, where p, q and r are integers.

(3)

(c) The point C has coordinates $(k, 0)$. The line AC is perpendicular to the line AB. Find the value of the constant k.

(3)

(Total 11 marks)

Q18.

A circle with centre C has equation $x^2 + y^2 + 14x - 10y + 49 = 0$.

(a) Express this equation in the form

$$(x - a)^2 + (y - b)^2 = r^2$$

(b) Write down:

(i) the coordinates of C;

(3)

(ii) the radius of the circle.

(2)

(c) Sketch the circle.

(2)

(d) A line has equation $y = kx + 6$, where k is a constant.

(i) Show that the x-coordinates of any points of intersection of the line and the circle satisfy the equation $(k^2 + 1)x^2 + 2(k + 7)x + 25 = 0$.

(2)

(ii) The equation $(k^2 + 1)x^2 + 2(k + 7)x + 25 = 0$ has equal roots. Show that

$$12k^2 - 7k - 12 = 0$$

(3)

(iii) Hence find the values of k for which the line is a tangent to the circle.

(2)

(Total 14 marks)

Q19.

A curve is defined by the parametric equations

$$x = 8t^2 - t, \qquad y = \frac{3}{t}$$

(a) Show that the cartesian equation of the curve can be written as $xy^2 + 3y = k$, stating the value of the integer k.

(2)

(b) (i) Find an equation of the tangent to the curve at the point P, where $t = \frac{1}{4}$.

(7)

(ii) Verify that the tangent at P intersects the curve when $x = \frac{3}{2}$.

(2)
(Total 11 marks)

Q20.

The line AB has equation $4x - 3y = 7$.

(a) (i) Find the gradient of AB.

(2)

(ii) Find an equation of the straight line that is parallel to AB and which passes through the point $C(3, -5)$, giving your answer in the form $px + qy = r$, where p, q and r are integers.

(3)

(b) The line AB intersects the line with equation $3x - 2y = 4$ at the point D. Find the coordinates of D.

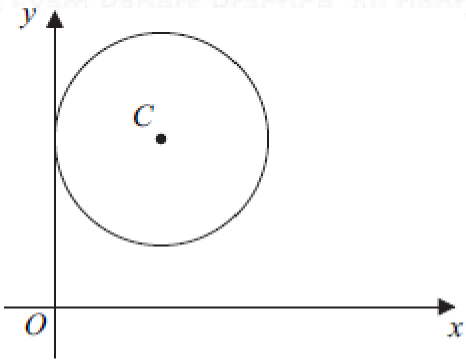
(3)

(c) The point E with coordinates $(k - 2, 2k - 3)$ lies on the line AB. Find the value of the constant k.

(2)
(Total 10 marks)

Q21.

The circle with centre $C(5, 8)$ touches the y-axis, as shown in the diagram.



(a) Express the equation of the circle in the form

$$(x - a)^2 + (y - b)^2 = k$$

(2)

(b) (i) Verify that the point $A(2, 12)$ lies on the circle.

(1)

(ii) Find an equation of the tangent to the circle at the point A, giving your answer in the form $sx + ty + u = 0$, where s, t and u are integers.

(5)

(c) The points P and Q lie on the circle, and the mid-point of PQ is M (7, 12).

(i) Show that the length of CM is $n\sqrt{5}$, where n is an integer.

(2)

(ii) Hence find the area of triangle PCQ.

(3)
(Total 13 marks)

Q22.

The gradient, $\frac{dy}{dx}$, of a curve C at the point (x, y) is given by

$$\frac{dy}{dx} = 20x - 6x^2 - 16$$

(a) (i) Show that y is increasing when $3x^2 - 10x + 8 < 0$.

(2)

(ii) Solve the inequality $3x^2 - 10x + 8 < 0$.

(4)

(b) The curve C passes through the point $P(2, 3)$.

(i) Verify that the tangent to the curve at P is parallel to the x-axis.

(2)

(ii) The point $Q(3, -1)$ also lies on the curve. The normal to the curve at Q and the tangent to the curve at P intersect at the point R. Find the coordinates of R.

(7)
(Total 15 marks)

Q23.

The line AB has equation $3x + 2y = 7$. The point C has coordinates $(2, -7)$.

(a) (i) Find the gradient of AB.

(2)

(ii) The line which passes through C and which is parallel to AB crosses the y-axis at the point D. Find the y-coordinate of D.

(3)

(b) The line with equation $y = 1 - 4x$ intersects the line AB at the point A. Find the coordinates of A.

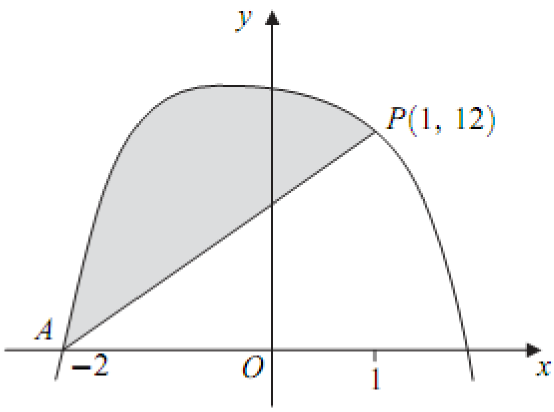
(3)

(c) The point E has coordinates $(5, k)$. Given that CE has length 5, find the two possible values of the constant k.

(3)
(Total 11 marks)

Q24.

The curve sketched below passes through the point A $(-2, 0)$.



The curve has equation $y = 14 - x - x^4$ and the point P $(1, 12)$ lies on the curve.

(a) (i) Find the gradient of the curve at the point P.

(3)

(ii) Hence find the equation of the tangent to the curve at the point P, giving your answer in the form $y = mx + c$.

(2)

(b) (i) Find $\int_{-2}^{1} (14 - x - x^4)\,dx$.

(5)

(ii) Hence find the area of the shaded region bounded by the curve $y = 14 - x - x^4$ and the line AP.

(2)
(Total 12 marks)

Q25.

A circle has centre C $(-3, 1)$ and radius $\sqrt{13}$.

(a) (i) Express the equation of the circle in the form

$$(x-a)^2+(y-b)^2=k$$

(2)

(ii) Hence find the equation of the circle in the form

$$x^2+y^2+mx+ny+p=0$$

where m, n and p are integers.

(3)

(b) The circle cuts the y-axis at the points A and B. Find the distance AB.

(3)

(c) (i) Verify that the point D (−5, −2) lies on the circle.

(1)

(ii) Find the gradient of CD.

(2)

(iii) Hence find an equation of the tangent to the circle at the point D.

(2)

(Total 13 marks)

Q26.

The line AB has equation $7x + 3y = 13$.

(a) Find the gradient of AB.

(2)

(b) The point C has coordinates (–1, 3).

(i) Find an equation of the line which passes through the point C and which is parallel to AB.

(2)

(ii) The point $\left(1\frac{1}{2}, -1\right)$ is the mid-point of AC. Find the coordinates of the point A.

(2)

(c) The line AB intersects the line with equation $3x + 2y = 12$ at the point B. Find the coordinates of B.

(3)

(Total 9 marks)

Q27.

A circle has centre C (3, –8) and radius 10.

(a) Express the equation of the circle in the form

$$(x - a)^2 + (y - b)^2 = k$$

(2)

(b) Find the x-coordinates of the points where the circle crosses the x-axis.

(3)

(c) The tangent to the circle at the point A has gradient $\frac{5}{2}$. Find an equation of the line CA, giving your answer in the form $rx + sy + t = 0$, where r, s and t are integers.

(3)

(d) The line with equation $y = 2x + 1$ intersects the circle.

(i) Show that the x-coordinates of the points of intersection satisfy the equation

$$x^2 + 6x - 2 = 0$$

(3)

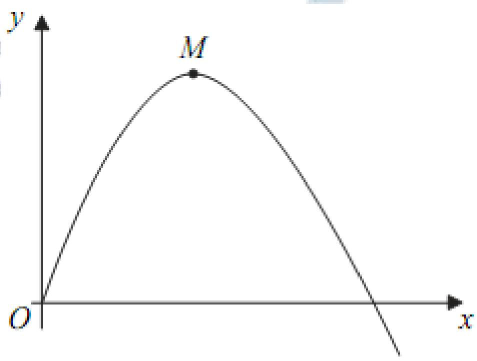
(ii) Hence show that the x-coordinates of the points of intersection are of the form $m \pm \sqrt{n}$, where m and n are integers.

(2)

(Total 13 marks)

Q28.

The diagram shows part of a curve with a maximum point M.



The curve is defined for $x \geq 0$ by the equation

$$y = 6x - 2x^{\frac{3}{2}}$$

(a) Find $\frac{\mathrm{d}y}{\mathrm{d}x}$.

(3)

(b) (i) Hence find the coordinates of the maximum point M.

(3)

(ii) Write down the equation of the normal to the curve at M.

(1)

(c) The point $P\left(\frac{9}{4}, \frac{27}{4}\right)$ lies on the curve.

(i) Find an equation of the normal to the curve at the point P, giving your answer in the form $ax + by = c$, where a, b and c are positive integers.

(4)

(ii) The normals to the curve at the points M and P intersect at the point R. Find the coordinates of R.

(2)

(Total 13 marks)

Q29.

The variables x and Y, where $Y = \log_{10} y$, are related by the equation

$$Y = mx + c$$

where m and c are constants.

(a) Given that $y = ab^x$, express a in terms of c, and b in terms of m.

(3)

(b) It is given that $y = 12$ when $x = 1$ and that $y = 27$ when $x = 5$.

On the diagram below, draw a linear graph relating x and Y.

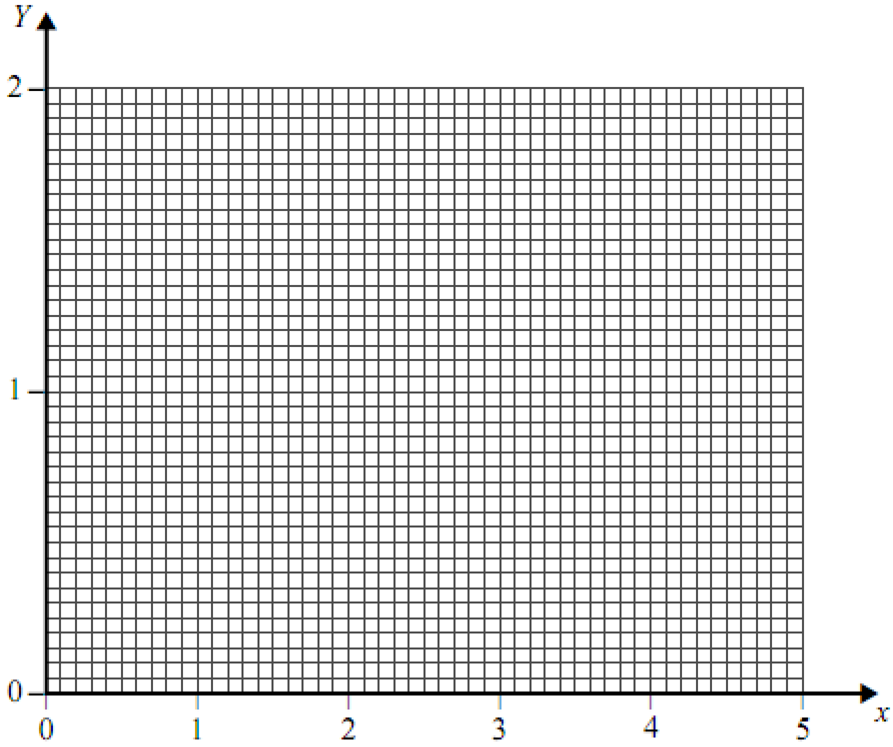
(3)

(c) Use your graph to estimate, to two significant figures:

(i) the value of y when $x = 3$;

(2)

(ii) the value of a.



(2)
(Total 10 marks)

Q30.

The triangle ABC has vertices $A(1, 3)$, $B(3, 7)$ and $C(-1, 9)$.

(a) (i) Find the gradient of AB.

(2)

(ii) Hence show that angle ABC is a right angle.

(2)

(b) (i) Find the coordinates of M, the mid-point of AC.

(2)

(ii) Show that the lengths of AB and BC are equal.

(3)

(iii) Hence find an equation of the line of symmetry of the triangle ABC.

(3)
(Total 12 marks)

Q31.

(a) Express $(x - 5)(x - 3) + 2$ in the form $(x - p)^2 + q$, where p and q are integers.

(3)

(b) (i) Sketch the graph of $y = (x - 5)(x - 3) + 2$, stating the coordinates of the minimum point and the point where the graph crosses the y-axis.

(3)

(ii) Write down an equation of the tangent to the graph of $y = (x - 5)(x - 3) + 2$ at its vertex.

(2)

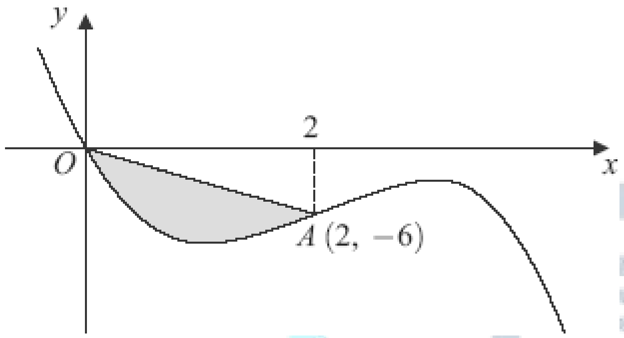
(c) Describe the geometrical transformation that maps the graph of $y = x^2$ onto the graph of $y = (x - 5)(x - 3) + 2$.

(3)

(Total 11 marks)

Q32.

The curve with equation $y = 12x^2 - 19x - 2x^3$ is sketched below.



The curve crosses the x-axis at the origin O, and the point A (2, –6) lies on the curve.

(a) (i) Find the gradient of the curve with equation $y = 12x^2 - 19x - 2x^3$ at the point A.

(4)

(ii) Hence find the equation of the normal to the curve at the point , giving your answer in the form $x + py + q = 0$, where p and q are integers.

(3)

(b) (i) Find the value of $\int_0^2 (12x^2 - 19x - 2x^3)\,dx$.

(5)

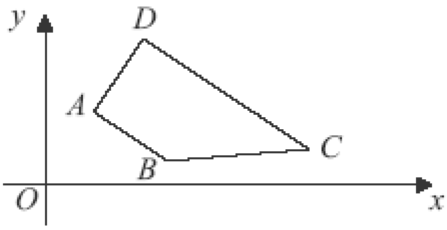
(ii) Hence determine the area of the shaded region bounded by the curve and the line OA.

(3)

(Total 15 marks)

Q33.

The trapezium $ABCD$ is shown below.



The line AB has equation $2x + 3y = 14$ and DC is parallel to AB.

(a) Find the gradient of AB.

(2)

(b) The point D has coordinates (3, 7).

(i) Find an equation of the line DC.

(2)

(ii) The angle BAD is a right angle. Find an equation of the line AD, giving your answer in the form $mx + ny + p = 0$, where m, n and p are integers.

(4)

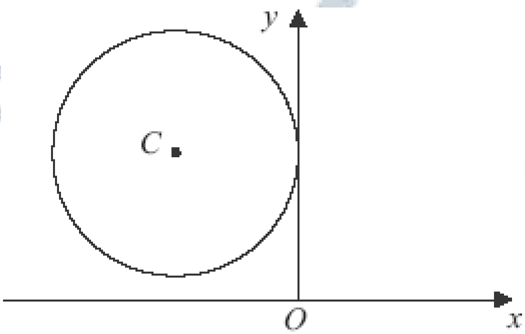
(c) The line BC has equation $5y - x = 6$. Find the coordinates of B.

(3)

(Total 11 marks)

Q34.

A circle with centre $C(-5, 6)$ touches the y-axis, as shown in the diagram.



(a) Find the equation of the circle in the form

$$(x - a)^2 + (y - b)^2 = r^2$$

(3)

(b) (i) Verify that the point $P(-2, 2)$ lies on the circle.

(1)

(ii) Find an equation of the normal to the circle at the point P.

(3)

(iii) The mid-point of PC is M. Determine whether the point P is closer to the point M or to the origin O.

(4)
(Total 11 marks)

Q35.

The points A and B have coordinates (1, 6) and (5, –2) respectively. The mid-point of AB is M.

(a) Find the coordinates of M.

(2)

(b) Find the gradient of AB, giving your answer in its simplest form.

(2)

(c) A straight line passes through M and is perpendicular to AB.

(i) Show that this line has equation $x - 2y + 1 = 0$.

(3)

(ii) Given that this line passes through the point $(k, k + 5)$, find the value of the constant k.

(2)
(Total 9 marks)